



Microprocessor & Computer Architecture (μ pCA)

UE24CS251B

Session
- 3

Microprocessor & Computer Architecture (μ pCA)



UE24CS251B

ARM program structure
and
Introduction to Arm
Simulator

Address	Instruction & Data
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```

        .text
Address of Instruction 1
ARM Instruction_1
Address of Instruction 2
ARM Instruction_2
.....
.....
.....
.....
Address of Instruction n
ARM Instruction_n

```

Note: Blue color depicts the code written by the programmer
Orange color depicts the address assigned during execution
Declaration of variable 1
Declaration of variable 2

General Format of Instruction

MNEMONIC {condition} {S} {Rd}, Operand1, Operand2

MNEMONIC - Short name of the instruction. *Eg: ADD, SUB...*

{condition} - Condition that is needed to be met in order for the instruction to be executed *Eg: EQ, NE, GT, LT, LE, AL*

{S} - An optional suffix. If S is specified, the condition flags are updated on the result of the operation. *Eg: To set N, Z, C, V of CPSR*

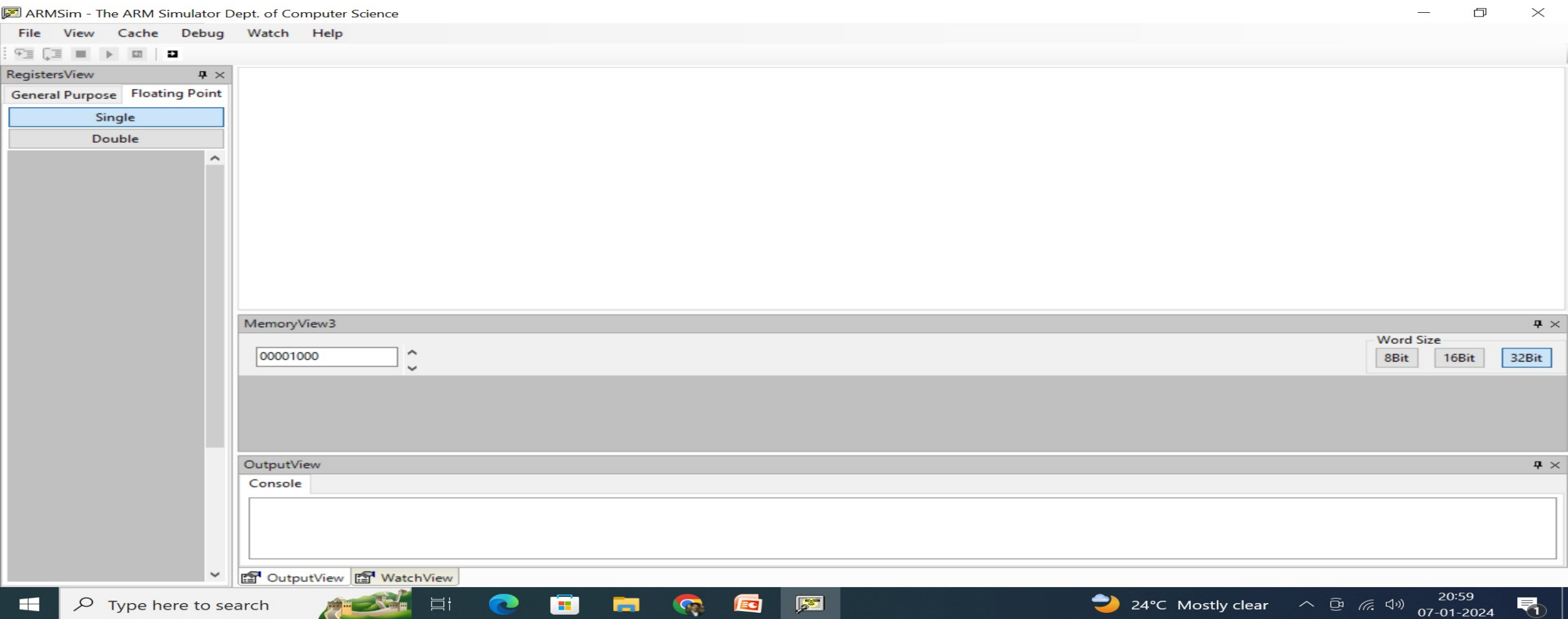
{Rd} - Register (destination) for storing the result of the instruction

Operand1 - First operand. Either a register or an immediate value

Operand2 - Second (flexible) operand. Can be an immediate value (number) or a register with an optional shift

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ARMSIMULATOR



NOTE: Recommended to use and demo on ARM simulator directly.

Loadi ng Sampl e Pr ogr am on ARMSI MULATOR



ARMSim - The ARM Simulator Dept. of Computer Science

File View Cache Debug Watch Help

RegistersView

General Purpose Floating Point

Hexadecimal

Unsigned Decimal

Signed Decimal

R0 : 00000000

R1 : 00000000

R2 : 00000000

R3 : 00000000

R4 : 00000000

R5 : 00000000

R6 : 00000000

R7 : 00000000

R8 : 00000000

R9 : 00000000

R10 (s1) : 00000000

R11 (fp) : 00000000

R12 (ip) : 00000000

R13 (sp) : 00005400

R14 (lr) : 00000000

R15 (pc) : 00001000

CPSR Register

Negative (N) : 0

Zero (Z) : 0

Carry (C) : 0

Overflow (V) : 0

IRQ Disable: 1

FIQ Disable: 1

Thumb (T) : 0

CPU Mode : System

0x000000df

addition.s

; Simple Code for Addition of 2 numbers.

00001000:E3A01003 MOV R1, #3

00001004:E3A02004 MOV R2, #4

00001008:E0813002 ADD R3, R1, R2

MemoryView3

00001000

Word Size

8Bit 16Bit 32Bit

00001000 E3A01003 E3A02004 E0813002 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181

0000102C 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181

00001058 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181

00001084 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181 81818181

OutputView

Console Stdin/Stdout/Stderr

Loading assembly language file F:\academics\subjects\MPCA\MPCA 2024\Slides\Sample Programs\addition.s

OutputView WatchView

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(upCA)

Register View (ARMSIM)

Simulation

ARMSim - The ARM Simulator Dept. of Computer Science

File View Cache Debug Watch Help

RegistersView

General Purpose Floating Point

Hexadecimal
Unsigned Decimal
Signed Decimal

R0 : 00000000
R1 : 00000000
R2 : 00000000
R3 : 00000000
R4 : 00000000
R5 : 00000000
R6 : 00000000
R7 : 00000000
R8 : 00000000
R9 : 00000000
R10 (s1) : 00000000
R11 (fp) : 00000000
R12 (ip) : 00000000
R13 (sp) : 00005400
R14 (lr) : 00000000
R15 (pc) : 00001000

CPSR Register
Negative (N) : 0
Zero (Z) : 0
Carry (C) : 0
Overflow (V) : 0
IRQ Disable : 1
FIQ Disable : 1
Thumb (T) : 0
CPU Mode : System

0x000000df

a1.s

```
00001000:E3A00003  mov r0, #3
00001004:E3A02004  Mov r2, #4
00001008:E0801002  ADD R1, R0, R2
```

MemoryView3

00001000

Word Size

8Bit 16Bit 32Bit

00001000	E3A00003	E3A02004	E0801002	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181
0000102C	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181
00001058	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181
00001084	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181

OutputView

Console Stdin/Stdout/Stderr

Loading assembly language file F:\academics\subjects\MPCA\MPCA 2024\a1.s

OutputView WatchView

General Format of Instruction Example

ARMSim - The ARM Simulator Dept. of Computer Science

FileViewCacheDebugWatchHelp

RegistersView

General PurposeFloating Point

Hexadecimal

Unsigned Decimal

Signed Decimal

R0 : 00000000

R1 : 00000003

R2 : 00000004

R3 : 00000007

R4 : 00000000

R5 : 00000000

R6 : 00000000

R7 : 00000000

R8 : 00000000

R9 : 00000000

R10 (s1) : 00000000

R11 (fp) : 00000000

R12 (ip) : 00000000

R13 (sp) : 00005400

R14 (lr) : 00000000

R15 (pc) : 0000100c

CPSR Register

Negative (N) : 0

Zero (Z) : 0

Carry (C) : 0

Overflow (V) : 0

IRQ Disable : 1

FIQ Disable : 1

Thumb (T) : 0

CPU Mode : System

0x000000df

addition.s

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0000102C	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181
00001058	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181
00001084	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181	81818181

OutputView

ConsoleStdin/Stdout/Stderr

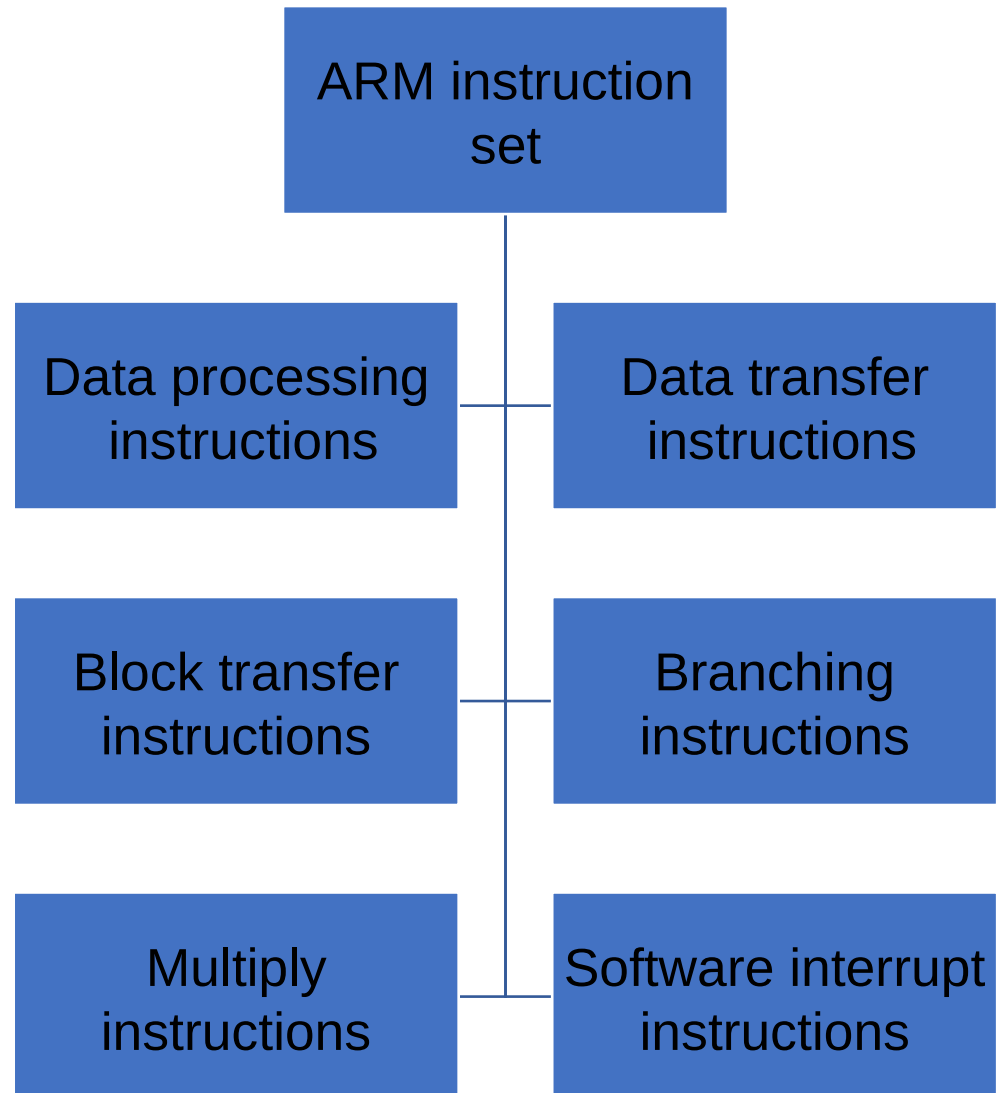
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Samples of following types of Numerical Data on Simulator

Demonstration of the following data is shown in the simulator.

- Decimal Numbers
- Hexadecimal Numbers
- Octal Numbers
- Usage of Bytes, Decimal and Hexadecimal data in the code and in the Simulator
- Read and observe the contents of the following in the simulator.
 - Memory
 - Stack
 - Registers
 - Flag Register
 - Console

ARM Instruction Set





THANK YOU



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Engineering